

Analysis PhD Exam, September 2010

Write solutions in a neat and logical fashion, giving complete reasons for all steps.

Attempt EIGHT problems.

1. Let $f \in L^1[0, 1]$. Suppose that

$$\int_0^1 x^n f(x) dx = 0 \quad \text{for all } n = 0, 1, 2, \dots$$

Prove that $f = 0$ almost everywhere. (Hint: reduce the problem to showing that if $\int_E f(x) dx = 0$ for every Borel E , then $f = 0$ a.e.)

2. Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space, and let $g : X \rightarrow \mathbb{C}$ be a measurable function. Prove that the operator $M_g : f \mapsto gf$ is bounded on $L^p(\mu)$ ($1 \leq p \leq \infty$) if and only if $g \in L^\infty(\mu)$, in which case $\|M_g\| = \|g\|_\infty$.

3. This problem has two parts.

- a) Define monotone class, and state the Monotone Class Theorem.
- b) State the Fubini–Tonelli theorem.

4. This problem has two parts.

- a) Let ν be a signed measure on $(X, \mathcal{M})$. Define what it means for a set to be positive, negative or null for ν , and state the Hahn decomposition theorem.
- b) Prove that if ν is a signed measure, then there exist unique positive measures ν^+, ν^- so that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$.

5. Let $(k_n)_{n=1}^\infty$ be a sequence of measurable, 1-periodic functions on $\mathbb{R}$ with the following properties:

Prove that if $f \in L^1[-\frac{1}{2}, \frac{1}{2}]$, then $f * k_n \rightarrow f$ in $L^1[-\frac{1}{2}, \frac{1}{2}]$.

(Here $*$ denotes convolution on $[-\frac{1}{2}, \frac{1}{2}]$: $(f * g)(x) = \int_{-1/2}^{1/2} f(t)g(x-t) dt$.)

- a) $k_n \geq 0$ a.e., for all n .
- b) $\int_{-1/2}^{1/2} k_n(t) dt = 1$ for all n .
- c) For each $\delta > 0$, $k_n \rightarrow 0$ uniformly on the set $\delta \leq |t| \leq \frac{1}{2}$.

A second page was presumably part of the original exam, but it was not present in the exam archive.